Multisensory Modalities beyond Vision

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1. Sound modality

Sound is a **mechanical vibration** that propagates through a medium (air, water, solid or fluid materials) and is ultimately **perceived by the human ear** directly as an auditory sensation. Sound vibrations generate **pressure waves** travelling outward from the source in all geometric directions. When the waves reach the human ear, they stimulate the **eardrum** and the **auditory system**, converting the mechanical energy into **neural signals** that the brain interprets as sound.

1.1. Modeling

A **sound source** produces vibrations travelling through a medium and are captured by a **microphone** → This device converts the mechanical vibrations into **electrical signals** → The signal is analyzed and transformed into digital form through two main computer operations: **Sampling** and **Quantization** → Sampling involves taking discrete measurements of the continuous sound wave at several regular intervals → Quantization assigns numerical values (codes) to the sampled points, enabling its storage and manipulation by the different digital systems.

1.2. Digital parameters

To reproduce sound accurately, the **sampling rate** must be at least twice the highest perceived frequency present in the sampled signal according to the **Nyquist theorem**. Since the human hearing range is [**20 Hz – 2 KHz**], the standard audio systems adopt a sampling rate of **44.1 kHz**. The **bit depth**, determines the number of bits we use to encode each sampled value. This number directly influence the precision and dynamic range of the recorded sound. Higher bit depth allows us for finer distinctions in amplitude, resulting in a smoother, more precise recordings with less quantization noise in the produced sound.

1.3. Sound utility

Sound contributes both in **semantic meaning** and **contextual richness** of the user experience:

- ➔ **Carrier of explicit information**: **Speech, dialogue, narration, music** (directly communicate messages, emotions, or narrative content). The mentioned auditory elements guide the user's understanding and enhance the expressiveness of multimedia presentations.
- ➔ **Contextual and environmental function**: Through **spatial cues** (directionality, distance, reverberation, and ambient noise). Sound helps audience perceiving **virtual spaces, realism, and VR immersion and presence**. Subtle auditory details, such as footsteps approaching from behind or fading echo in a large hall, also provide the brain with necessary information about the context and environment contributing to the overall sense of presence.

1.4. Advanced tools

- ➔ **Recording and playback systems** capture sound waves using microphones or audio sensors and reproduce them through speakers ensuring that the original acoustic properties are preserved as faithfully as possible. Beyond simple sound recording.
- ➔ **Synthesis and transformation tools** (Digital synthesizers, samplers, and audio processors) allow the generation of entirely new original sounds or the alteration of existing ones, providing composers and designers with large creative possibilities and choices.

Available on zenodo – DOI : 10.5281/zenodo.17379102

- ➔ **Sound effects (SFX)** tools are used for enhancing **realism** or evoke **specific emotions** in multimedia productions, like the sound of footsteps in a movie or the different simulated ambiances of virtual environments. **Following this,** e
- ➔ **Editing and montage** enable accurate temporal and structural ordering of the audio tracks to ensure the wright synchronization with visual elements or narrative cues.
- ➔ **Mixing and mastering** processes adjust sound levels, equalize frequencies, and apply compression or spatial effects to attain clarity, depth, and a professional finish suitable for end-user public distribution.

1.5. Implementation

Sound poses more significant **storage and processing demands** than text modality, owing to its continuous nature and high data rate.

For example, a single minute of uncompressed high-quality audio require several megabytes of storage, which historically limited its use in early digital systems.

Sound remains one of the **most pervasive and powerful channels** in all multimedia applications, distinguished by its **expressive**, **immersive**, and **emotive** characteristics. Unlike other modalities, it instantly conveys mood, urgency, or atmosphere, enhancing comprehension and engagement.

The development of **advanced compression techniques** (**MP3 – MPEG-1 Audio Layer III** and **AAC – Advanced Audio Coding**) has revolutionized sound storage and transmission reducing file sizes dramatically and eliminating inaudible frequencies and redundant data while maintaining perceptual audio quality through robust psychoacoustic representations. In combination with **efficient encoding and streaming methods**, these advanced tools have made the **diffusion of audio content** (Digital music, podcasts, online education…) feasible on a large global scale.

2. Haptic modality

This modality is intricate and multifaceted. Unlike sound/image primarily processed through distinct sensory organs (ear/eye), Haptic engages the **entire body** as a perceptual interface. It integrates **cutaneous sensations** (touch, pressure, temperature, and texture) and **proprioceptive feedback** (Awareness of body position, posture, orientation, and movement). This multifaced nature makes haptic perception **rich and complex** Combining external contact with internal bodily awareness.

Cognitive perspective: Haptic perception emerges through the **integration of somatic stimuli** (Skin receptors, muscles, and joints) with **audiovisual information** processed by the brain. The resulting multimodal synthesis creates a **unified haptic sensory experience** allowing humans to interact with their environment in a realistic and embodied way. The haptic sense is thus essential for **spatial understanding**, **object manipulation**, and also **emotional engagement**, making a bridge between physical interaction and cognitive interpretation.

Example: In a **VR** medical training environment, haptic feedback devices (**force-feedback gloves** or **surgical simulators)** allow users to really feel tissue resistance, instrument vibrations, and pressure changes in medical acts. These tactile cues, combined with visual and auditory elements, create realistic learning experience enhancing **manual precision**, **spatial awareness**, and **sensorimotor coordination** that replicates the conditions of real surgical practice.

3. Olfaction and Gustation

☑ What we can call **chemical senses** (**olfaction or smell** and **gustation or taste**) play an important role in human perception and contribute to survival and the richness of sensory human experience.
→ These two modalities are closely interrelated
→ They constitute the foundation of **flavor perception**, influencing memory, emotion, and behavior.

Olfaction is the biological process by which the human detects and identifies odors – **Gustation** complements the olfaction by recognizing **chemical substances dissolved in food or drink**.

3.1. Biological mechanism

First, the **biological mechanism** of **Olfaction** (the **Sense of Smell)**:

→ Starts in **olfactory epithelium** – Thin, yellowish membrane high within the human nasal cavity.

→ This epithelium contains millions of **olfactory receptor neurons** specialized to bind with specific **airborne chemical molecules**.

→ When these molecules are inhaled, they interact with the receptors, initiating a cascade of **electrochemical signals** that travel to the **olfactory bulb** (a neural structure responsible for the **initial processing and organization** of olfactory information).

→ The signals are then conveyed to higher brain regions, including the **olfactory cortex** and the **limbic system**, where they are allied with memory and emotion.

→ This is why certain smells can evoke vivid recollections or emotional states.

Second, the **biological mechanism** of **Gustation** (**Sense of Taste)**:

→ The **taste buds**, located on the **tongue**, **soft palate**, and **oral mucosa**, contain clusters of specialized **gustatory receptor cells** that respond to the different dissolved molecules.

→ They detect the five primary taste known qualities: **sweet, sour, salty, bitter, and umami.**

→ Each quality provides critical information about the nutritional or toxic value of what is ingested.

→ The sensory information is then transmitted through the **gustatory nerves** to the **gustatory cortex**, where the brain interprets these signals and produces the conscious experience of taste.

3.2. Utility

Olfaction is rarely implemented in current multimedia systems although possesses an exceptional ability to **stimulate memory and emotion**. The **olfactory system's direct connection to the limbic structures** of the brain allows scents to activate vivid recollections and emotional responses more effectively than traditional visual or auditory cues. In VR/AR contexts, olfactory feedback might significantly **strengthen the presence sense**, making simulated environments feel more authentic and more psychologically engaging.

*For instance, the scent of sea salt in a coastal VR simulation or the aroma of wood in a medieval setting **anchor users more deeply** into the virtual experience, fostering immersion with **contextual realism** and **emotional resonance**.*

Gustation remains less explored due to **technical and physiological complexity and constraints**. Its potential applications are recognized in specialized medical and cognitive fields. **Culinary simulations** and **gastronomy training platforms** could use digital taste interfaces to teach flavor composition and palate development. In **medical rehabilitation**, gustatory feedback might assist patients recovering from taste loss or eating disorders via reactivating sensory pathways. Furthermore, in **experimental entertainment and multisensory storytelling**, taste cues could be synchronized with narrative or visual rendering to expand the expressive range of numerical media to create **novel aesthetic and emotional experiences**.

Example – Immersive dining experiences: VR headsets are combined with scent diffusers and flavor-modulating devices. Participants might visually perceive a tropical forest, smell exotic fruit aromas, and taste a corresponding dish (all aligned to the storyline). Such multisensory coherence illustrates how olfaction and gustation can **transform traditional media into embodied immersive experiences**, bridging the gap between perception, emotion, and technology.

3.3. Human capacity

Humans can perceive **five universally recognized basic tastes**:

1. **Sweet**: For sugars and energy-rich foods;
2. **Sour**: Related to acids and the balance of pH;
3. **Salty**: For mineral salts and electrolyte regulation;

4. **Bitter**: Serving as a natural warning signal for potentially toxic substances;
5. **Umami**: Savory derived from amino acids (glutamate) commonly found in protein-rich foods.

In the **olfaction** realm, humans are remarkably sensitive to around **10,000** distinct odors, and research suggests that olfactory system may distinguish up to **one trillion olfactory stimuli**. This extraordinary capacity underlines **human olfactory perception richness and complexity** and emphasizes how taste and smell together shape the experiences of flavor, emotion, and memory.

3.4. Challenges and limitations

➔ Perceptual and Physical Constraints

Smell and taste lack a standardized perceptual scale or measurable perceptual range, which makes it difficult to reproduce or precisely control and imitate the corresponding sensations. The **chemical synthesis, diffusion, and storage of odors** require compounds that are both **stable and non-toxic**, adding more complexity. Moreover, environmental factors such as ambient humidity, temperature, and airflow can significantly alter the perceived intensity and quality of a scent.

➔ Ergonomic Constraints

Integrating olfactory and gustatory feedback into multimedia applications raises important and conflictual ergonomic issues. **Artificial scents** may interfere with or contradict other modality stimuli **(**visual and auditory), potentially disrupting the sense of immersion. Furthermore, **taste and smell delivery systems** must be accurately controlled and highly localized (particularly within the **tongue or nasal cavity**) to prevent user discomfort, cross-contamination, and overstimulation.

➔ Cognitive Constraints

The **persistence and slow diffusion of odors and taste** pose a major challenge for temporal synchronization with the other sensory modalities. Once released, smells and taste linger in the environment or the mouth and are difficult to remove quickly, which complicates **real-time adaptation and interactivity**. Misalignment between olfactory, visual, and auditory cues can lead to **perceptual contradictions**, reducing realism and confusing the user's sensory immersive experience.

3.5. Advanced Tools and Research

➔ Advances in Olfactory Simulation

Recent research has led to the emergence of what we can call **olfactory displays**, specialized devices capable of emitting **controlled and programmable scents**. Complementing these are **micro-diffusers** allowing **targeted and localized scent delivery**, enhancing realism and user immersion. Furthermore, **wearable olfactory systems** are increasingly being integrated into **VR** and **AR** applications, enabling the synchronization of smell with visual and auditory cues.

Experimental research is also expanding in **cross-modal perception**, investigating how olfactory stimuli interact with other sensory modalities (vision, sound, and haptic) to create richer and more coherent multisensory experiences.

➔ Advances in Gustatory Simulation

The gustatory dimension has seen the rise of **electronic tongues**: Peripherals capable of **analyzing, recognizing, and replicating taste profiles** through advanced sensing solutions. Researchers are also exploring **electrostimulation techniques** applied directly to the **taste buds** to evoke or modulate taste sensations without chemical substances or direct end-contact. This advancement is paving the way for **wearable gustatory interfaces** designed for **VR and AR environments** and offering new opportunities for **culinary education, sensory rehabilitation, and immersive entertainment** through the digital recreation of taste experiences.